

dsbda-lab-ass-2

February 14, 2025

1 Assignment 2

1.1 Import Required Library

```
[3]: import pandas as pd
```

1.2 Read CSV file into Dataframe

```
[6]: df = pd.read_csv('StudentDatabase.csv')
```

```
[8]: df
```

```
[8]:
```

	Roll No.	Name	City	Age	Marks	Grades
0	1	Aarav Deshmukh	Pune	15.0	25.0	E
1	2	Aishwarya Patil	Sangli	16.0	45.0	C
2	3	Rahul Joshi	Satara	14.0	33.0	D
3	4	Shruti Kulkarni	Nashik	15.0	66.0	A
4	5	Tanmay Shinde	Nashik	16.0	56.0	B
5	6	Priya Pawar	Nashik	15.0	NaN	NaN
6	7	Siddharth Gaikwad	Nashik	15.0	35.0	D
7	8	Ananya Narkar	Nashik	16.0	12.0	F
8	9	Rohan Jadhav	Pune	NaN	33.0	D
9	10	Pooja Thakur	Sangli	15.0	NaN	NaN
10	11	Aditya Chavan	Pune	16.0	66.0	A
11	12	Kavita Taware	Sangli	15.0	60.0	A
12	13	Karan Gawade	Nashik	NaN	67.0	A
13	14	Snehal Rajurkar	NaN	16.0	45.0	C
14	15	Vikram Borse	Pune	16.0	63.0	A
15	16	Mansi Mahajan	Nashik	NaN	44.0	C
16	17	Devendra Sawant	Sangli	16.0	34.0	D
17	18	Sanya Koli	NaN	15.0	45.0	C
18	19	Abhishek Gite	Nashik	15.0	23.0	E
19	20	Smita Sawant	Nashik	NaN	13.0	F
20	21	Nikhil Phadke	NaN	15.0	54.0	B
21	22	Medha Khandekar	Sangli	15.0	28.0	E
22	23	Shubham Waghmare	NaN	NaN	39.0	D
23	24	Neha Joshi	Pune	14.0	11.0	F

24	25	Vinayak Mane	Pune	16.0	7.0	F
25	26	Shital Deshpande	Nashik	NaN	25.0	E
26	27	Ameya Bhosale	Sangli	15.0	67.0	A
27	28	Kiran Karande	Sangli	NaN	22.0	E
28	29	Sushant Shirodkar	Pune	15.0	45.0	C
29	30	Radhika Yadav	Pune	15.0	23.0	E

1.3 Check Structure of Dataframe

```
[11]: df.shape
```

```
[11]: (30, 6)
```

```
[13]: df.size
```

```
[13]: 180
```

```
[15]: df.ndim
```

```
[15]: 2
```

1.4 Check Attribute of Dataframe

```
[18]: df.columns
```

```
[18]: Index(['Roll No.', 'Name', 'City', 'Age', 'Marks', 'Grades'], dtype='object')
```

1.5 Check the data types of columns in a DataFrame.

```
[21]: df.dtypes
```

```
[21]: Roll No.      int64
      Name        object
      City        object
      Age         float64
      Marks       float64
      Grades      object
      dtype: object
```

1.6 Check Attribute info of Dataframe

```
[24]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 6 columns):
```

#	Column	Non-Null Count	Dtype
0	Roll No.	30 non-null	int64
1	Name	30 non-null	object
2	City	26 non-null	object
3	Age	23 non-null	float64
4	Marks	28 non-null	float64
5	Grades	28 non-null	object

dtypes: float64(2), int64(1), object(3)
memory usage: 1.5+ KB

1.7 Check Description of Dataframe

```
[27]: df.describe()
```

```
[27]:
```

	Roll No.	Age	Marks
count	30.000000	23.000000	28.000000
mean	15.500000	15.260870	38.785714
std	8.803408	0.619192	18.623675
min	1.000000	14.000000	7.000000
25%	8.250000	15.000000	24.500000
50%	15.500000	15.000000	37.000000
75%	22.750000	16.000000	54.500000
max	30.000000	16.000000	67.000000

1.8 Check null values of Dataframe

```
[30]: df.isnull().any()
```

```
[30]:
```

Roll No.	False
Name	False
City	True
Age	True
Marks	True
Grades	True

dtype: bool

```
[32]: df.isnull().sum()
```

```
[32]:
```

Roll No.	0
Name	0
City	4
Age	7
Marks	2
Grades	2

dtype: int64

1.9 Drop all rows that having no values

```
[35]: df.dropna()
```

```
[35]:
```

	Roll No.	Name	City	Age	Marks	Grades
0	1	Aarav Deshmukh	Pune	15.0	25.0	E
1	2	Aishwarya Patil	Sangli	16.0	45.0	C
2	3	Rahul Joshi	Satara	14.0	33.0	D
3	4	Shruti Kulkarni	Nashik	15.0	66.0	A
4	5	Tanmay Shinde	Nashik	16.0	56.0	B
6	7	Siddharth Gaikwad	Nashik	15.0	35.0	D
7	8	Ananya Narkar	Nashik	16.0	12.0	F
10	11	Aditya Chavan	Pune	16.0	66.0	A
11	12	Kavita Taware	Sangli	15.0	60.0	A
14	15	Vikram Borse	Pune	16.0	63.0	A
16	17	Devendra Sawant	Sangli	16.0	34.0	D
18	19	Abhishek Gite	Nashik	15.0	23.0	E
21	22	Medha Khandekar	Sangli	15.0	28.0	E
23	24	Neha Joshi	Pune	14.0	11.0	F
24	25	Vinayak Mane	Pune	16.0	7.0	F
26	27	Ameya Bhosale	Sangli	15.0	67.0	A
28	29	Sushant Shirodkar	Pune	15.0	45.0	C
29	30	Radhika Yadav	Pune	15.0	23.0	E

1.10 Fill NA values with mean and mode values

```
[38]: df1=df.copy()
for col in df.columns:
    if df1[col].isnull().any():
        if df1[col].dtype == 'object': # For categorical (non-numeric) columns
            df1[col] = df1[col].fillna(value=df1[col].mode()[0])
        else: # For numeric columns
            df1[col] = df1[col].fillna(value=df1[col].mean())

# Check for remaining missing values
df1.isnull().sum()
df1
```

```
[38]:
```

	Roll No.	Name	City	Age	Marks	Grades
0	1	Aarav Deshmukh	Pune	15.000000	25.000000	E
1	2	Aishwarya Patil	Sangli	16.000000	45.000000	C
2	3	Rahul Joshi	Satara	14.000000	33.000000	D
3	4	Shruti Kulkarni	Nashik	15.000000	66.000000	A
4	5	Tanmay Shinde	Nashik	16.000000	56.000000	B
5	6	Priya Pawar	Nashik	15.000000	38.785714	A
6	7	Siddharth Gaikwad	Nashik	15.000000	35.000000	D
7	8	Ananya Narkar	Nashik	16.000000	12.000000	F

8	9	Rohan Jadhav	Pune	15.26087	33.000000	D
9	10	Pooja Thakur	Sangli	15.00000	38.785714	A
10	11	Aditya Chavan	Pune	16.00000	66.000000	A
11	12	Kavita Taware	Sangli	15.00000	60.000000	A
12	13	Karan Gawade	Nashik	15.26087	67.000000	A
13	14	Snehal Rajurkar	Nashik	16.00000	45.000000	C
14	15	Vikram Borse	Pune	16.00000	63.000000	A
15	16	Mansi Mahajan	Nashik	15.26087	44.000000	C
16	17	Devendra Sawant	Sangli	16.00000	34.000000	D
17	18	Sanya Koli	Nashik	15.00000	45.000000	C
18	19	Abhishek Gite	Nashik	15.00000	23.000000	E
19	20	Smita Sawant	Nashik	15.26087	13.000000	F
20	21	Nikhil Phadke	Nashik	15.00000	54.000000	B
21	22	Medha Khandekar	Sangli	15.00000	28.000000	E
22	23	Shubham Waghmare	Nashik	15.26087	39.000000	D
23	24	Neha Joshi	Pune	14.00000	11.000000	F
24	25	Vinayak Mane	Pune	16.00000	7.000000	F
25	26	Shital Deshpande	Nashik	15.26087	25.000000	E
26	27	Ameya Bhosale	Sangli	15.00000	67.000000	A
27	28	Kiran Karande	Sangli	15.26087	22.000000	E
28	29	Sushant Shiroadkar	Pune	15.00000	45.000000	C
29	30	Radhika Yadav	Pune	15.00000	23.000000	E

1.11 Fill NA values with forward fill values

```
[41]: df2=df.copy()
      for col in df2.columns:
          if df2[col].isnull().any():
              df2[col] = df2[col].ffill()
df2.isnull().sum()
df2
```

```
[41]:
```

	Roll No.	Name	City	Age	Marks	Grades
0	1	Aarav Deshmukh	Pune	15.0	25.0	E
1	2	Aishwarya Patil	Sangli	16.0	45.0	C
2	3	Rahul Joshi	Satara	14.0	33.0	D
3	4	Shruti Kulkarni	Nashik	15.0	66.0	A
4	5	Tanmay Shinde	Nashik	16.0	56.0	B
5	6	Priya Pawar	Nashik	15.0	56.0	B
6	7	Siddharth Gaikwad	Nashik	15.0	35.0	D
7	8	Ananya Narkar	Nashik	16.0	12.0	F
8	9	Rohan Jadhav	Pune	16.0	33.0	D
9	10	Pooja Thakur	Sangli	15.0	33.0	D
10	11	Aditya Chavan	Pune	16.0	66.0	A
11	12	Kavita Taware	Sangli	15.0	60.0	A
12	13	Karan Gawade	Nashik	15.0	67.0	A
13	14	Snehal Rajurkar	Nashik	16.0	45.0	C

14	15	Vikram Borse	Pune	16.0	63.0	A
15	16	Mansi Mahajan	Nashik	16.0	44.0	C
16	17	Devendra Sawant	Sangli	16.0	34.0	D
17	18	Sanya Koli	Sangli	15.0	45.0	C
18	19	Abhishek Gite	Nashik	15.0	23.0	E
19	20	Smita Sawant	Nashik	15.0	13.0	F
20	21	Nikhil Phadke	Nashik	15.0	54.0	B
21	22	Medha Khandekar	Sangli	15.0	28.0	E
22	23	Shubham Waghmare	Sangli	15.0	39.0	D
23	24	Neha Joshi	Pune	14.0	11.0	F
24	25	Vinayak Mane	Pune	16.0	7.0	F
25	26	Shital Deshpande	Nashik	16.0	25.0	E
26	27	Ameya Bhosale	Sangli	15.0	67.0	A
27	28	Kiran Karande	Sangli	15.0	22.0	E
28	29	Sushant Shirodkar	Pune	15.0	45.0	C
29	30	Radhika Yadav	Pune	15.0	23.0	E

1.12 Fill NA values with backward fill values

```
[44]: df3=df.copy()
      for col in df3.columns:
          if df3[col].isnull().any():
              df3[col] = df3[col].bfill()
df3.isnull().sum()
df3
```

```
[44]:
```

	Roll No.	Name	City	Age	Marks	Grades
0	1	Aarav Deshmukh	Pune	15.0	25.0	E
1	2	Aishwarya Patil	Sangli	16.0	45.0	C
2	3	Rahul Joshi	Satara	14.0	33.0	D
3	4	Shruti Kulkarni	Nashik	15.0	66.0	A
4	5	Tanmay Shinde	Nashik	16.0	56.0	B
5	6	Priya Pawar	Nashik	15.0	35.0	D
6	7	Siddharth Gaikwad	Nashik	15.0	35.0	D
7	8	Ananya Narkar	Nashik	16.0	12.0	F
8	9	Rohan Jadhav	Pune	15.0	33.0	D
9	10	Pooja Thakur	Sangli	15.0	66.0	A
10	11	Aditya Chavan	Pune	16.0	66.0	A
11	12	Kavita Taware	Sangli	15.0	60.0	A
12	13	Karan Gawade	Nashik	16.0	67.0	A
13	14	Snehal Rajurkar	Pune	16.0	45.0	C
14	15	Vikram Borse	Pune	16.0	63.0	A
15	16	Mansi Mahajan	Nashik	16.0	44.0	C
16	17	Devendra Sawant	Sangli	16.0	34.0	D
17	18	Sanya Koli	Nashik	15.0	45.0	C
18	19	Abhishek Gite	Nashik	15.0	23.0	E
19	20	Smita Sawant	Nashik	15.0	13.0	F

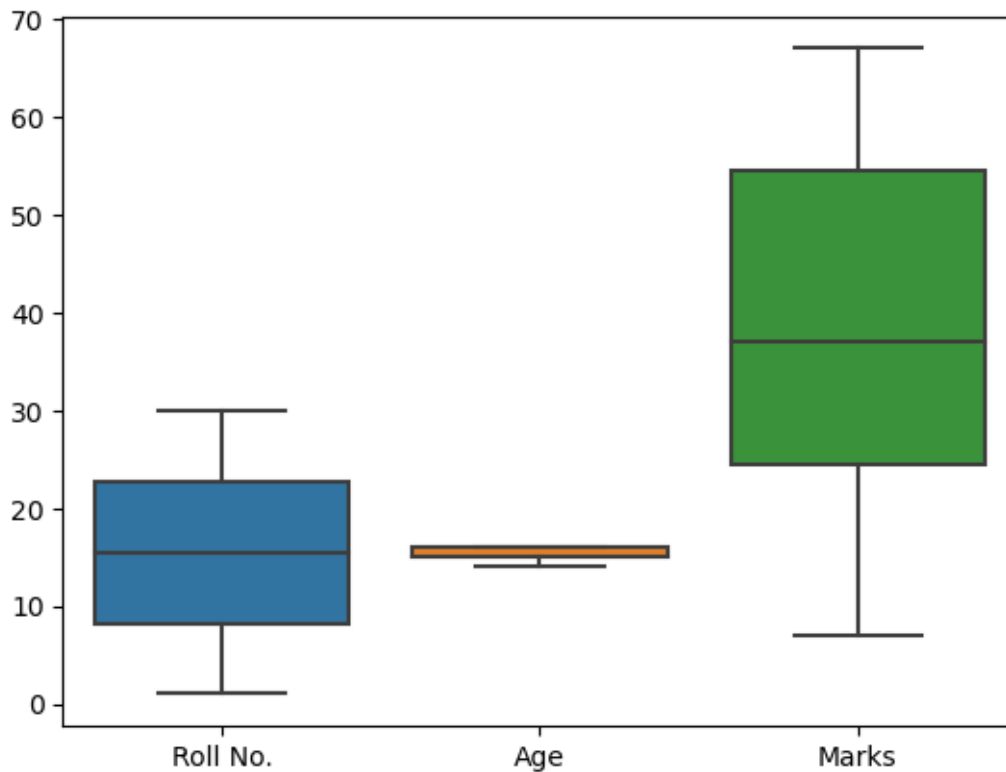
20	21	Nikhil Phadke	Sangli	15.0	54.0	B
21	22	Medha Khandekar	Sangli	15.0	28.0	E
22	23	Shubham Waghmare	Pune	14.0	39.0	D
23	24	Neha Joshi	Pune	14.0	11.0	F
24	25	Vinayak Mane	Pune	16.0	7.0	F
25	26	Shital Deshpande	Nashik	15.0	25.0	E
26	27	Ameya Bhosale	Sangli	15.0	67.0	A
27	28	Kiran Karande	Sangli	15.0	22.0	E
28	29	Sushant Shirodkar	Pune	15.0	45.0	C
29	30	Radhika Yadav	Pune	15.0	23.0	E

1.13 Visualize Data for better understanding

```
[47]: import seaborn as sns
import matplotlib.pyplot as plt
```

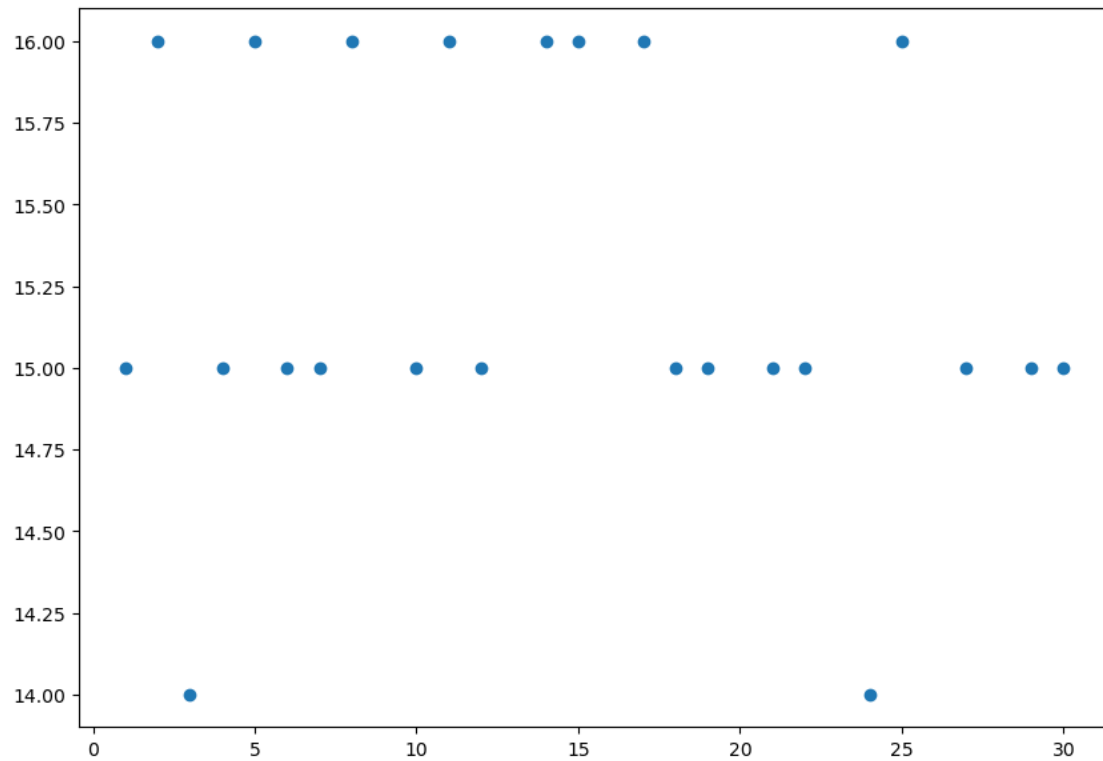
```
[48]: sns.boxplot(data=df)
```

```
[48]: <Axes: >
```

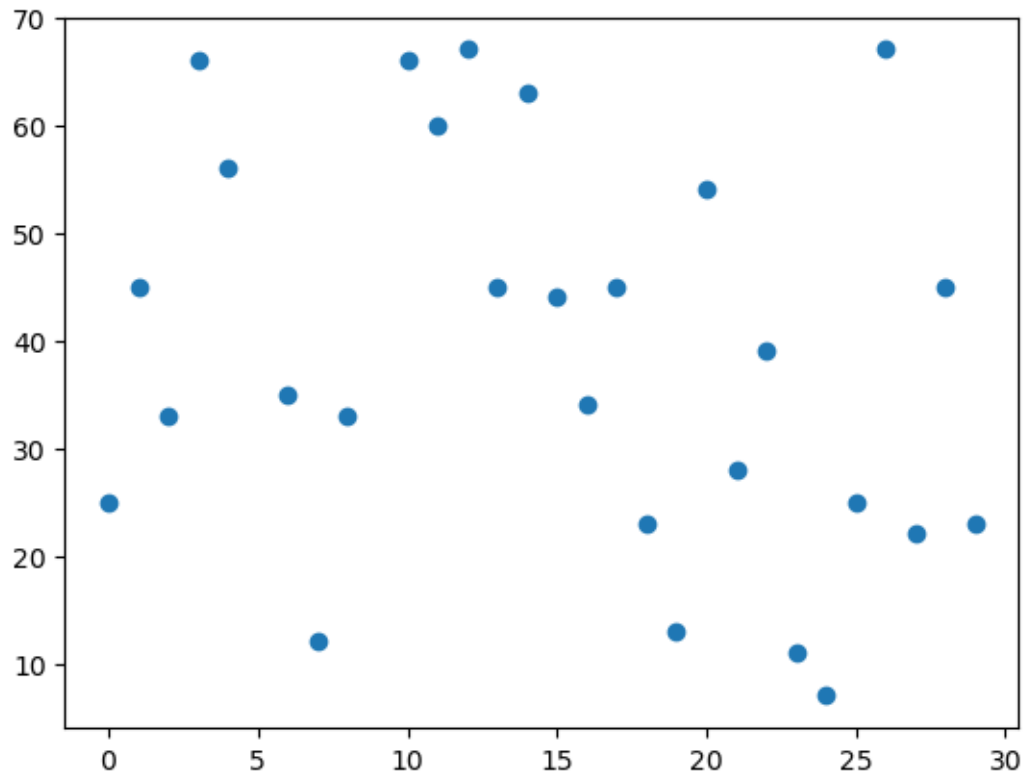


```
[51]: plt.figure(figsize=(10,7))  
plt.scatter(x=df['Roll No.'],y=df['Age'])
```

```
[51]: <matplotlib.collections.PathCollection at 0x1f42fa97910>
```



```
[53]: plt.scatter(x=df.index,y=df['Marks'])  
plt.show()
```

```
[55]: plt.figure(figsize=(12,6))
sns.barplot(data=df,x=df['Roll No.'],y=df['Marks'])
plt.title("Students marks")
plt.show()
```

